



**CHAITANYA BHARATHI
INSTITUTE OF TECHNOLOGY**
Autonomous Institute | Affiliated to Osmania University

CASE STUDY

Swiggy Delivery Time Prediction

Applying Predictive Analytics to Forecast Food-Delivery Times Using Random Forest Regression

Subject Applied Predictive Analytics

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The Problem — and the Solution Implemented

! Problem

Delivery times on Swiggy swing widely order-to-order, driven by factors that are invisible to the customer at checkout:

- Distance, traffic and weather vary a lot between orders
- Rider availability and multiple simultaneous deliveries add delay
- Customers see generic ETAs, causing complaints when orders run late
- Operations has no early signal to identify at-risk orders

Net effect: *inaccurate ETAs and delayed intervention hurt customer trust and operational efficiency.*

✓ Solution Implemented

A predictive-analytics pipeline was built to forecast delivery time up front and surface risk before it happens:

- Engineered distance, time and delay features from raw order data
- Trained a Random Forest Regressor to predict delivery time (min)
- Classified each order into a Low / Medium / High delay-risk band
- Published predictions to a Power BI dashboard for live monitoring

Outcome: *accurate, explainable ETAs and an early-warning signal ops can act on.*



Swiggy & the Delivery-Time Problem

● Company

Swiggy — one of India's largest online food-delivery and restaurant-discovery platforms, processing millions of orders daily across cities.

9K

ORDERS ANALYZED

● Business Problem

Delivery times are highly variable — driven by traffic, weather, distance and rider factors. Swiggy needed a way to predict how long a delivery will actually take, so customers see accurate ETAs and operations can flag orders at risk of delay.

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PREDICTIVE FEATURES

● Data Used

Structured, tabular order-level data: rider age & rating, restaurant/delivery GPS coordinates, order & pickup timestamps, weather, traffic density, vehicle type, city type and festival flag.

Structured

TABULAR ORDER DATA

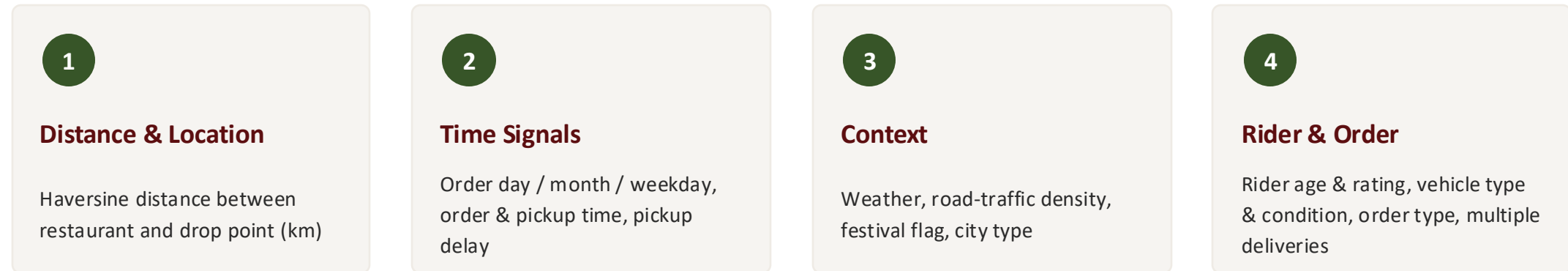


What Was Predicted — and How

Prediction target: *Time_taken (min)* — the actual delivery duration for each order, framed as a regression problem.



Key Predictive Features



Model: *Random Forest Regressor (100 trees) trained on an 80/20 train-test split, with categorical variables converted using Label Encoding.*



Results and Business Impact

9K

TOTAL ORDERS SCORED

26.2

AVG. ACTUAL DELIVERY (MIN)

26.3

AVG. PREDICTED DELIVERY (MIN)

8.4%

ORDERS FLAGGED HIGH-RISK

How the Model Was Evaluated

MAE & RMSE quantify the average error (in minutes) between predicted and actual delivery time.

R² Score measures how much of the variation in delivery time the model explains.

Feature Importance ranked Distance, Road Traffic Density and Pickup Delay as the strongest drivers of delivery time.

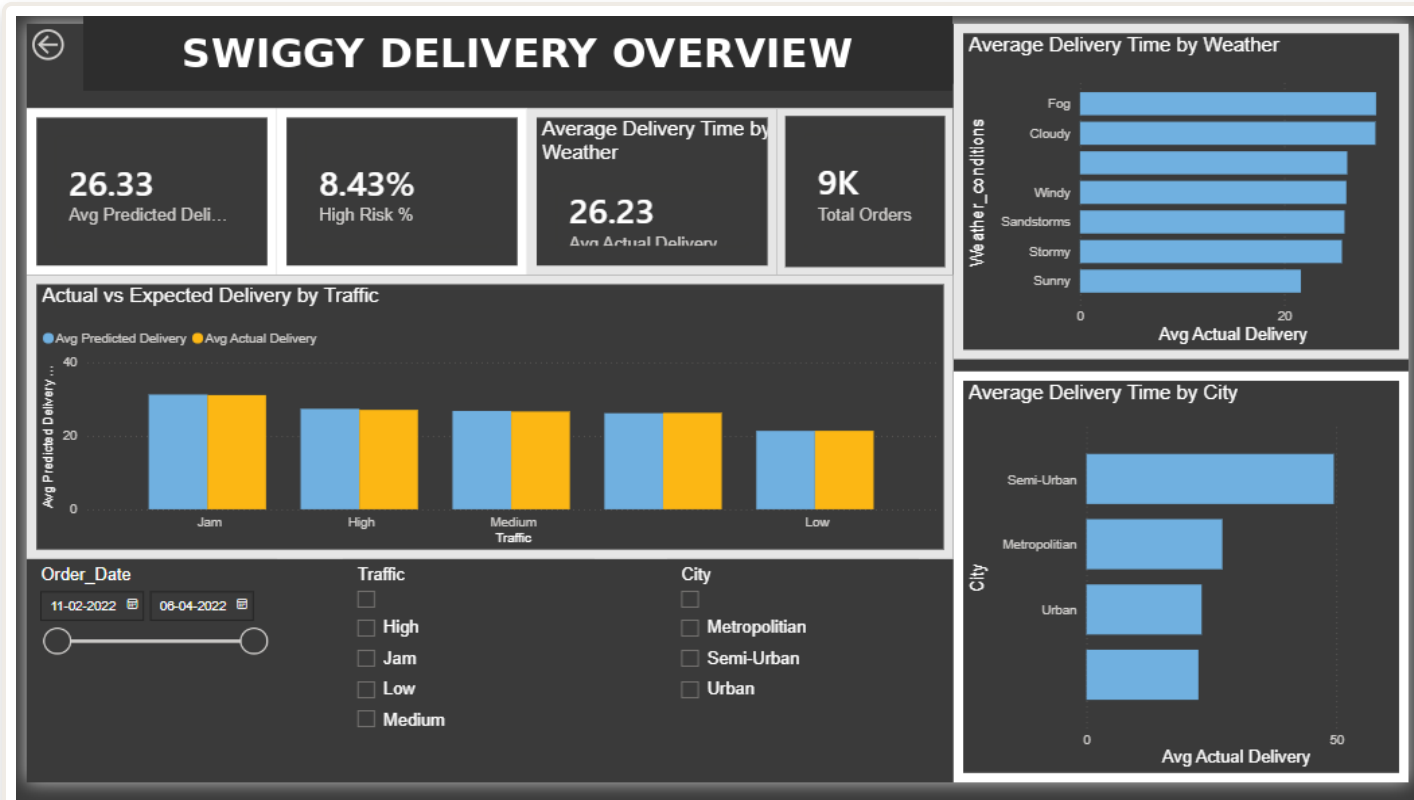
Business Impact

- More accurate ETAs shown to customers, reducing complaints
- Delay-risk flag lets ops teams intervene before an order is late
- Traffic/weather-driven delay patterns surfaced by city and route
- Insights feed directly into a live Power BI dashboard for monitoring



Bringing It Together in Power BI

Model predictions were exported to CSV and visualized in an interactive Power BI report for delivery-time monitoring.



Dashboard Highlights

- KPI cards: total orders, avg. actual vs predicted time, high-risk %
- Predicted vs actual delivery time compared across traffic levels
- Average delivery time broken down by weather and by city type
- Filters for order date, traffic and city enable drill-down analysis



Predictive Analytics, BI, Statistics & Data Mining

Applied to this project: statistics summarized the data, data mining and BI explored patterns, and predictive analytics (Random Forest) forecast delivery time.

